Parallel Lines: Extension

Name _____ Date _____ No calculator Version 1

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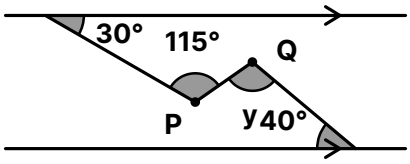
KEY WORDS parallel *(whakarara)* · alternate *(koki pārewa)* · co-interior *(koki roto-tahi)*

A The double zigzag ●●○

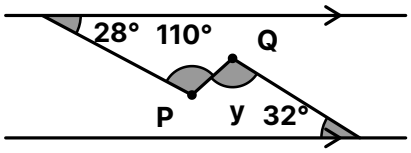
Remember: Draw a parallel line through each bend. Every angle at a bend splits into two alternate angles. Work from the angles you know to the ones you do not.

1. The two lines with arrows are parallel. Find the angle marked y. Show every step.

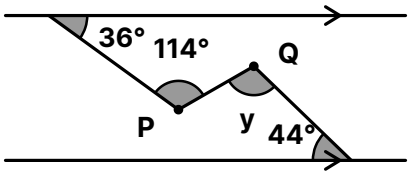
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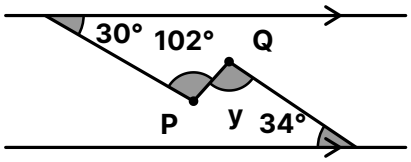
b)



c)



d)

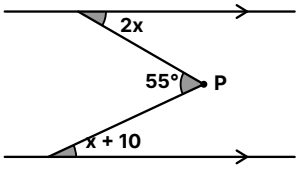


B Equations from the figure ●●●

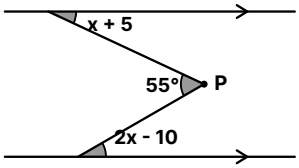
Remember: Decide first which rule links the marked angles (equal, add to 180°, or add to the third angle). Write that as an equation in x, solve it, then find every angle.

2. The two lines with arrows are parallel. Write an equation, solve it, and find each angle.

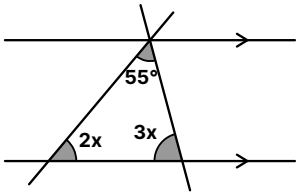
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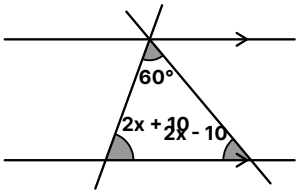
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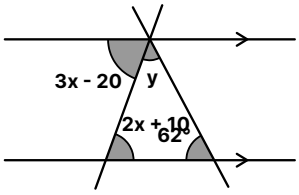
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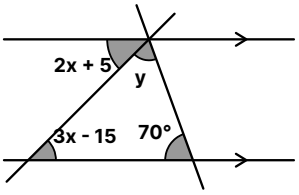
d)



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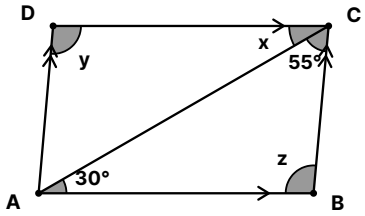
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C Diagonals ●●●

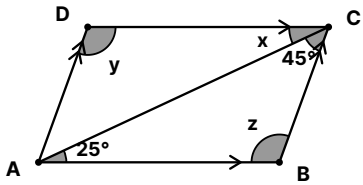
Remember: A diagonal is a line crossing two parallel sides, so it makes alternate angles with them. Then use the angle sum of the triangle it creates.

3. In each figure, AB is parallel to DC. Find the angles marked with letters. Give reasons.

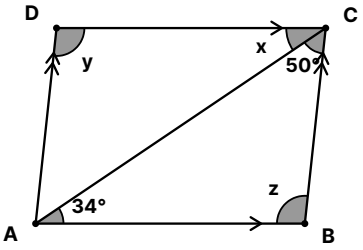
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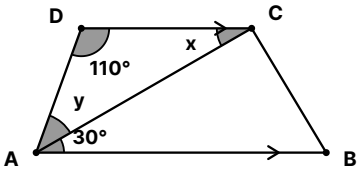
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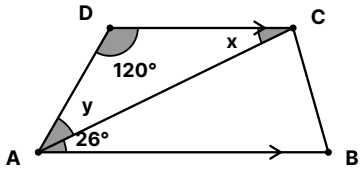
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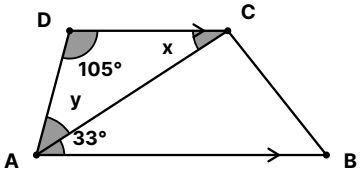
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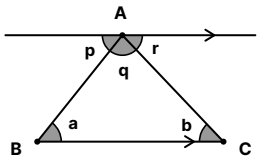
f)



D Proofs ●●●

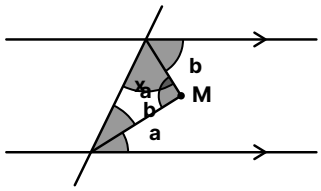
Watch out: A proof uses letters, not numbers, so it works for every triangle or every pair of lines. Give a reason for each line of the argument.

4. The line through A is parallel to BC. Use the figure to prove that the angles of a triangle add to 180°.



a) Write the proof.

5. The two lines with arrows are parallel. The co-interior angles at the two crossings are bisected (cut in half), and the bisectors meet at M.



a) Prove that the angle x at M is a right angle.

★ Extension

6. ABCD is a parallelogram with the diagonal AC. Angle BAC is 40° and angle ACB is 52°.

a) Without a figure, find all four angles of the parallelogram. Explain each step.

b) Explain why the diagonal AC can only bisect angle A when the parallelogram is a rhombus.

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Parallel Lines: Extension

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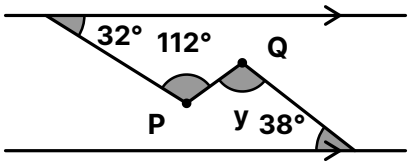
KEY WORDS parallel *(whakarara)* · alternate *(koki pārewa)* · co-interior *(koki roto-tahi)*

A The double zigzag ●●○

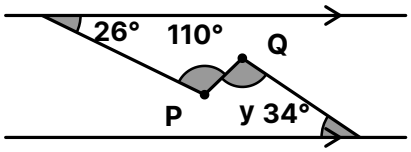
Remember: Draw a parallel line through each bend. Every angle at a bend splits into two alternate angles. Work from the angles you know to the ones you do not.

1. The two lines with arrows are parallel. Find the angle marked y. Show every step.

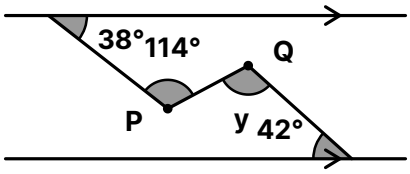
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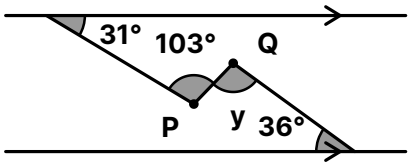
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c)



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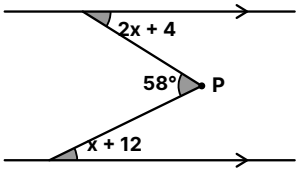


B Equations from the figure ●●●

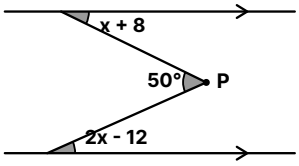
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2. The two lines with arrows are parallel. Write an equation, solve it, and find each angle.

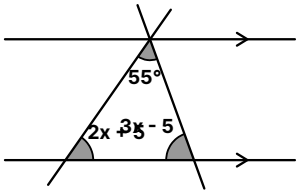
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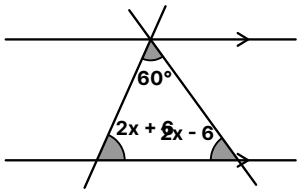
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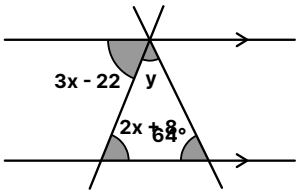
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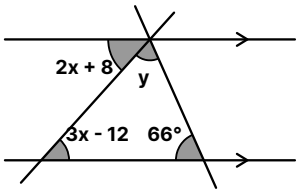
d)



e)



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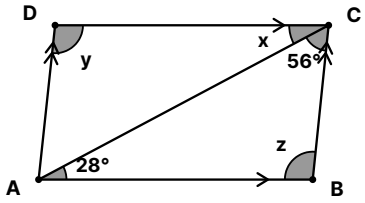
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C Diagonals ●●●

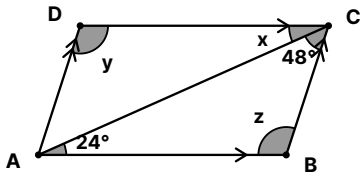
Remember: A diagonal is a line crossing two parallel sides, so it makes alternate angles with them. Then use the angle sum of the triangle it creates.

3. In each figure, AB is parallel to DC. Find the angles marked with letters. Give reasons.

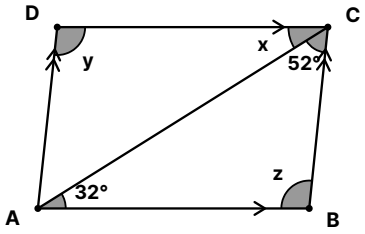
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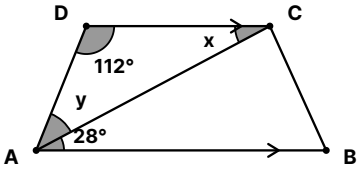
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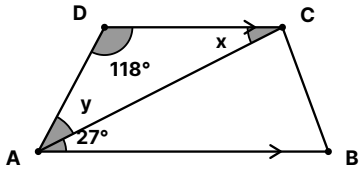
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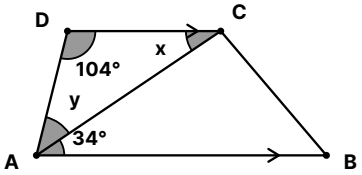
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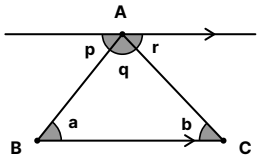
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D Proofs ●●●

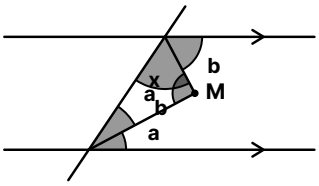
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4. The line through A is parallel to BC. Use the figure to prove that the angles of a triangle add to 180°.



a) Write the proof.

5. The two lines with arrows are parallel. The co-interior angles at the two crossings are bisected (cut in half), and the bisectors meet at M.



a) Prove that the angle x at M is a right angle.

Extension

6. ABCD is a parallelogram with the diagonal AC. Angle BAC is 38° and angle ACB is 54°.

a) Without a figure, find all four angles of the parallelogram. Explain each step.

b) Explain why the diagonal AC can only bisect angle A when the parallelogram is a rhombus.

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